Year 12 Maths - Similar to June 2024 Question 9

Similar questions - Pure & Statistics

Two new questions on the same topic as the exam question, a little more open-ended. They are not past exam questions. Answers are at the end.

1 A cube has volume $8a^{\frac{9}{2}}$ cm^3, where $a > 0$.

(a) Show that the length of an edge of the cube is $2a^{\frac{3}{2}}$ cm. **[2]**

(b) Show that the total surface area of the cube is $24a^3$ cm^2. **[2]**

(c) Given that the total surface area is 3 cm^2, find the value of a. **[2]**

2 The number of cells in a culture is modelled by $N = 5 \times 2^{t/3}$, where t is the time in hours after the start.

(a) Find the value of N when $t = 0$ and when $t = 6$. **[2]**

(b) Show that $N(t+3) = 2N(t)$, and explain what this tells you about the culture. **[2]**

(c) Find the value of t when $N = 40$. **[2]**

Question 1

(a) Edge $= \left(8a^{9/2}\right)^{1/3} = 8^{1/3} \times a^{3/2} = 2a^{3/2}$

(b) Surface area $= 6 \times \left(2a^{3/2}\right)^2 = 6 \times 4a^3 = 24a^3$

(c) $24a^3 = 3 \Rightarrow a^3 = \dfrac{1}{8} \Rightarrow a = \dfrac{1}{2}$

Question 2

(a) $N(0) = 5$ and $N(6) = 5 \times 2^2 = 20$

(b) $N(t+3) = 5 \times 2^{(t+3)/3} = 5 \times 2^{t/3} \times 2 = 2N(t)$: the number of cells doubles every 3 hours

(c) $5 \times 2^{t/3} = 40 \Rightarrow 2^{t/3} = 8 = 2^3 \Rightarrow t = 9$ hours